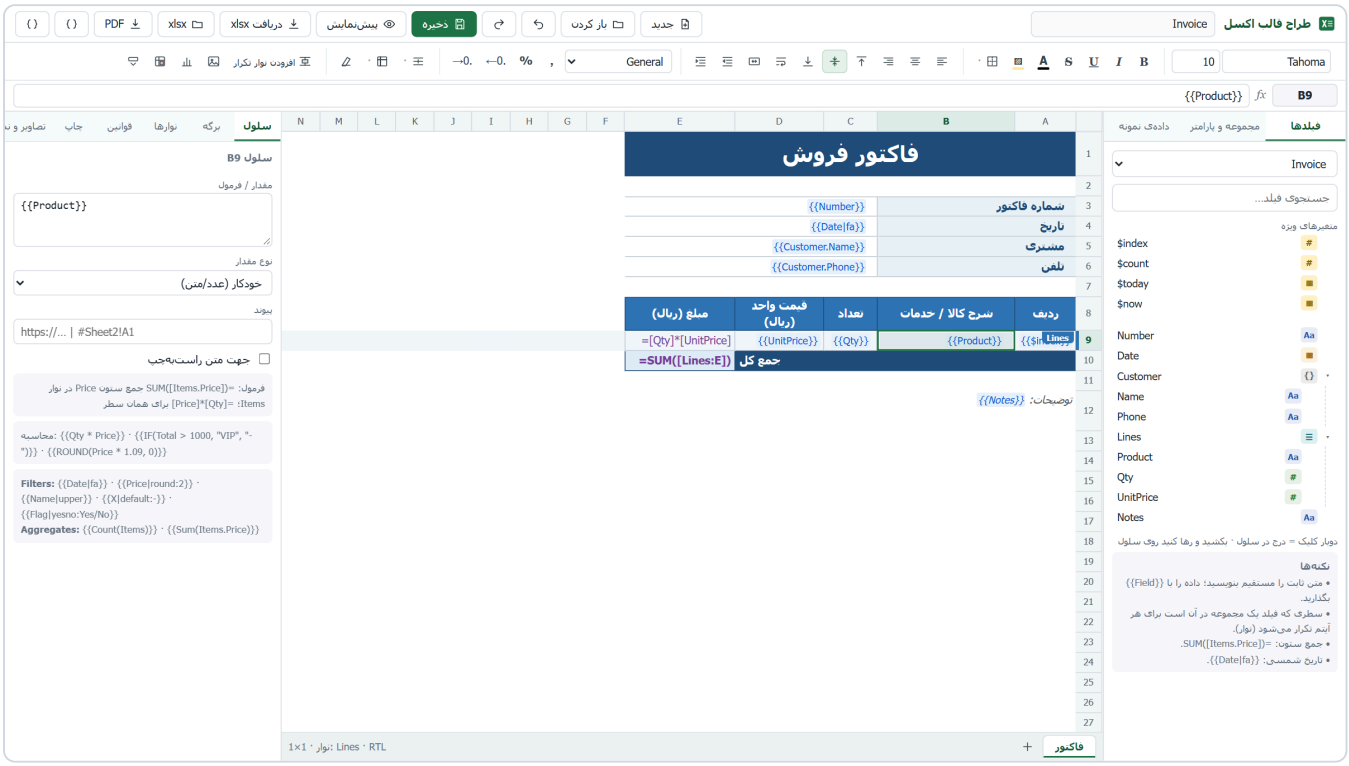
KSC.Excel — Complete Guide

A visual Excel/PDF template designer and report engine for .NET and .NET Framework

Design a template once in a visual designer; your code only supplies data and gets a beautiful `.xlsx` (or PDF) back — no EPPlus / ClosedXML / OpenXML SDK, no Excel installed on the server, no CDN (works on an intranet).

```
return this.ExcelFile("Invoice", model, "invoice.xlsx");                    // Excel
return File(svc.RenderPdf("Invoice", model), "application/pdf", "invoice.pdf");  // PDF
```



> Persian guide: KSC.Excel-Guide.fa.md · Architecture & design: ../KSC.Excel-Design.md (Persian)

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1. What is it?

The problem. Producing a good-looking Excel file in an enterprise app usually means picking a library, writing dozens of formatting lines per report, and redeploying for every colour change.

The solution.

Part	Description
Visual designer	Excel-like grid with a data panel, repeating bands, conditional formats, charts and live preview; Persian/English UI, right-to-left sheets
Engine	JSON template + data → `.xlsx` / PDF; no spreadsheet library dependency
Hosts	ASP.NET Core (8/10), ASP.NET MVC 5 (.NET Framework 4.7.2+), Windows desktop app (.NET 10 and .NET Framework 4.8)
Display	In-page HTML viewer (`KscExcelViewer`), **in-page PDF viewer** (`KscPdfViewer` / `Html.KscPdfViewer`) or file download

Packages

Package	Target	Purpose
`KSC.Excel`	`netstandard2.0`	Core: template model, render engine, XLSX writer, xlsx importer, HTML preview, designer API + embedded UI
`KSC.Excel.AspNetCore`	`net8.0`, `net10.0`	`AddKscExcel`, `MapKscExcelDesigner`, `ExcelFile`, `KscExcelViewer`

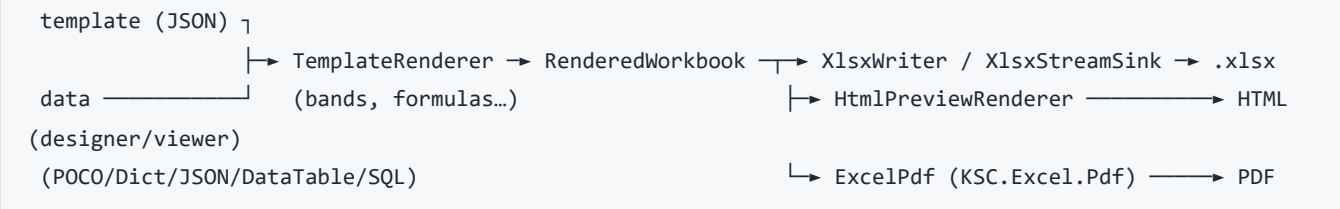
Package	Target	Purpose
`KSC.Excel.Mvc`	`net472`, `net48`	`KscExcel.Initialize`, designer route, `ExcelResult`, `Html.KscExcelViewer`
`KSC.Excel.Pdf`	`netstandard2.0`	PDF with correct Persian/Arabic shaping (SkiaSharp + HarfBuzz); works on .NET Framework 4.6.1+ **and** .NET 8/10
`KSC.Excel.Designer.Windows`	`net10.0-windows`, `net48`	Windows desktop designer (WPF + WebView2); one source, two runtimes

The core's only dependency is `Newtonsoft.Json`.

Key features

- **Data binding:** `{{Field}}`, calculated `{{Qty * Price}}`, filters (`|fa` Persian date, `|round`, …), aggregates, special variables.
- **Structure:** repeating bands and nested bands, sort & group with group totals, **cross-tab** (dynamic columns), sheet repeat, report sections (page header/report header/footer), **group wizard**.
- **File:** live formulas (with cached values), merges, freeze panes, auto-filter, conditional formats (7 kinds), data validation, images (static/dynamic), **charts** (column, bar, line, area, pie), hyperlinks, print setup.
- **External data:** read-only parameterised **SQL**, **REST/JSON endpoints**, **code providers** (EF/Dapper/any service), inline JSON, report parameters, custom functions/filters.
- **Reports like a band designer: barcodes & QR codes**, **page break per group**, **watermark**, conditional formats in PDF, rotated text.
- **Scale: streaming writer** (1 M rows ≈ 12 s, ≈ 47 MB peak memory).
- **More: import an existing xlsx**, **PDF + in-page PDF viewer**, Windows desktop app.
- **Quality:** output verified in real Excel; tests run on .NET 10 and .NET Framework 4.8.

Architecture at a glance



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2. Quick start: how do I open the designer?

The designer is a **web page** served by your app (or shown in a desktop window). Four ways:

A) Run the bundled sample (fastest)

```
git clone <repo-url>
cd KSC
dotnet run --project samples/KSC.Excel.Sample.Web --urls http://localhost:5199 -- --environment
Development
```

Open in Chrome/Edge:

```
http://localhost:5199/excel-designer
```

> Without `--environment Development` the designer requires a signed-in user and returns 401 to anonymous visitors — this is intentional (see Security).

The sample's home page links to the designer, the in-page viewer (`/Home/Viewer`), the Excel download (`/Home/Download`) and PDF (`/Home/Pdf`).

B) Inside your ASP.NET Core app

```
builder.Services.AddKscExcel();
var app = builder.Build();
app.MapKscExcelDesigner("/excel-designer").RequireAuthorization();   // always behind login
```

Then browse to `https://your-app/excel-designer` .

C) Inside your .NET Framework (MVC 5) app

```
// Global.asax.cs
KscExcel.Initialize(o => o.TemplateDirectory = "ExcelTemplates");
// RouteConfig.cs
routes.MapKscExcelDesigner();      // /excel-designer
```

D) Windows desktop app (no browser, no web site)

```
dotnet publish src/KSC.Excel.Designer.Windows -c Release -f net10.0-windows -r win-x64 --self-
contained false -o artifacts/KSC.Excel.Designer      # .NET 10 Desktop Runtime
dotnet publish src/KSC.Excel.Designer.Windows -c Release -f net48 -o artifacts/KSC.Excel.Designer-
net48                              # nothing to install on Windows 10/11 + WebView2
artifacts\KSC.Excel.Designer\KSC.Excel.Designer.exe
```

Details: section 9.

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3. Installation & requirements

Item	Requirement
.NET	SDK 8 or 10 to build; matching runtime to run
.NET Framework	4.7.2 / 4.8 for the MVC adapter (core works from 4.6.1)
Designer browser	Recent Chrome/Edge/Firefox
Excel	**Not required.** Files were verified in desktop Excel 2016+; LibreOffice / Excel Online / Google Sheets were not tested
Desktop app	Windows 10/11, .NET 10 Desktop Runtime, WebView2 Runtime (preinstalled on Windows 11)
PDF on Linux	A Persian font (e.g. Vazirmatn) and `SkiaSharp.NativeAssets.Linux.NoDependencies`

Add to your project

Until the NuGet packages are published, reference the projects directly:

```
<ItemGroup>
  <ProjectReference Include="..\KSC.Excel.AspNetCore\KSC.Excel.AspNetCore.csproj" />
  <ProjectReference Include="..\KSC.Excel.Pdf\KSC.Excel.Pdf.csproj" />   <!-- optional: PDF -->
</ItemGroup>
```

After publishing to NuGet: `dotnet add package KSC.Excel.AspNetCore` (and `KSC.Excel.Pdf`).

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4. Designer tour

4.1 Screen layout

Area	Contents
Top bar	template name · **New** · **Open** · Undo/Redo · **Save** · **Preview** · **Download xlsx** · import xlsx · **PDF** (when `UsePdf()` is on) · export/import JSON
Ribbon	font & size · bold/italic/underline/strike · text & fill colour · borders · horizontal/vertical align · wrap · merge · indent · number format · structure (insert/delete rows & columns) · freeze · clear format · **add repeat band** · image · chart · conditional format · validation
Formula bar	cell address · `fx` · value/formula

Area	Contents
Left panel	**Fields** · **Data sets & parameters** · **Sample data**
Grid	Excel-like cells; bands shown as coloured tags beside the row numbers
Right panel	**Cell** · **Sheet** · **Bands** · **Rules** · **Print** · **Images & charts**
Bottom	sheet tabs · selection status

4.2 Keyboard shortcuts

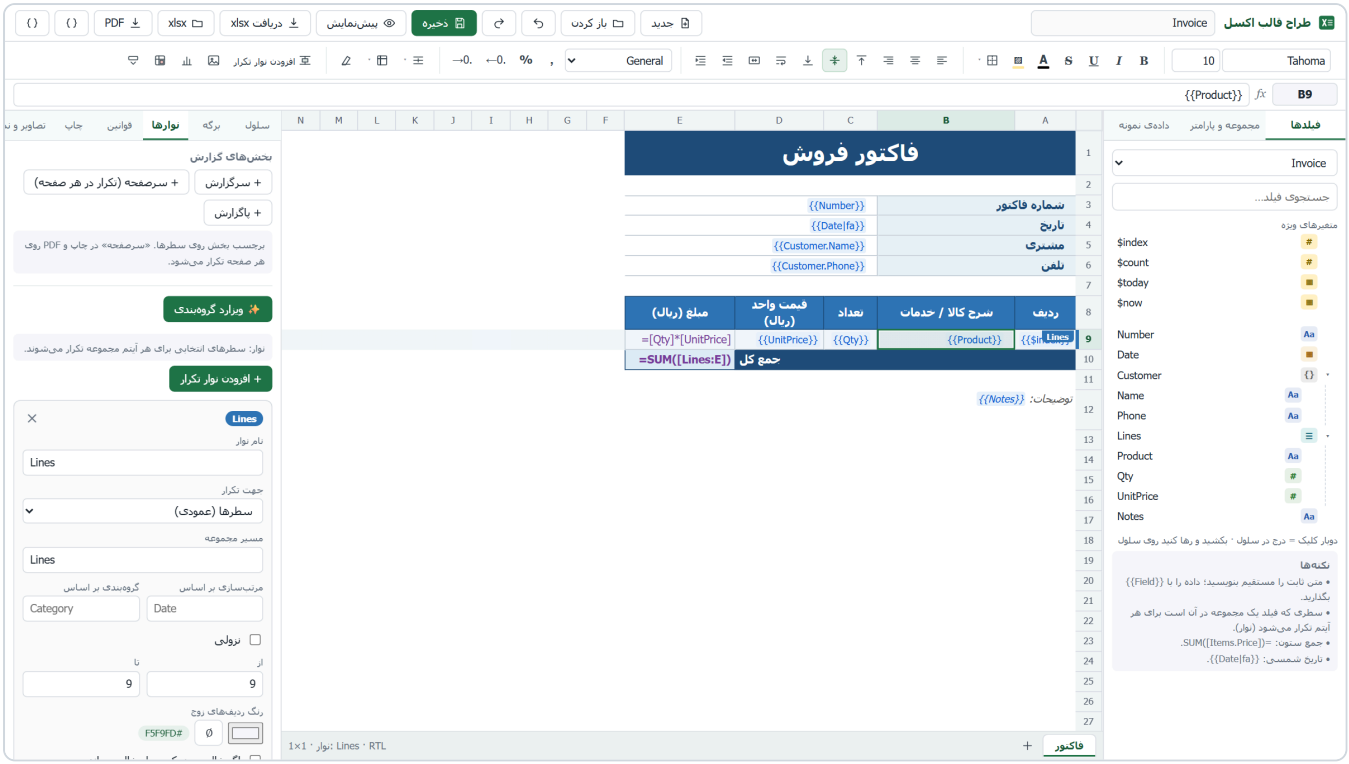
Key	Action	Key	Action
`Ctrl+S`	Save	`F2` / `Enter`	Edit cell
`Ctrl+Z` / `Ctrl+Y`	Undo / Redo	`Alt+Enter`	New line inside a cell
`Ctrl+B / I / U`	Bold / Italic / Underline	`Tab` / `Shift+Tab`	Next / previous cell
`Ctrl+C / X / V`	Copy / Cut / Paste (also from/to Excel)	`Delete`	Clear contents
`Ctrl+A`	Select all	`Ctrl+Arrow`	Jump to the data edge
`Shift+Click/Arrow`	Extend selection	Right-click	Cell/row/column menu

In a right-to-left sheet the Right arrow moves to the previous column (as in Excel). Drag a header edge to resize a column/row.

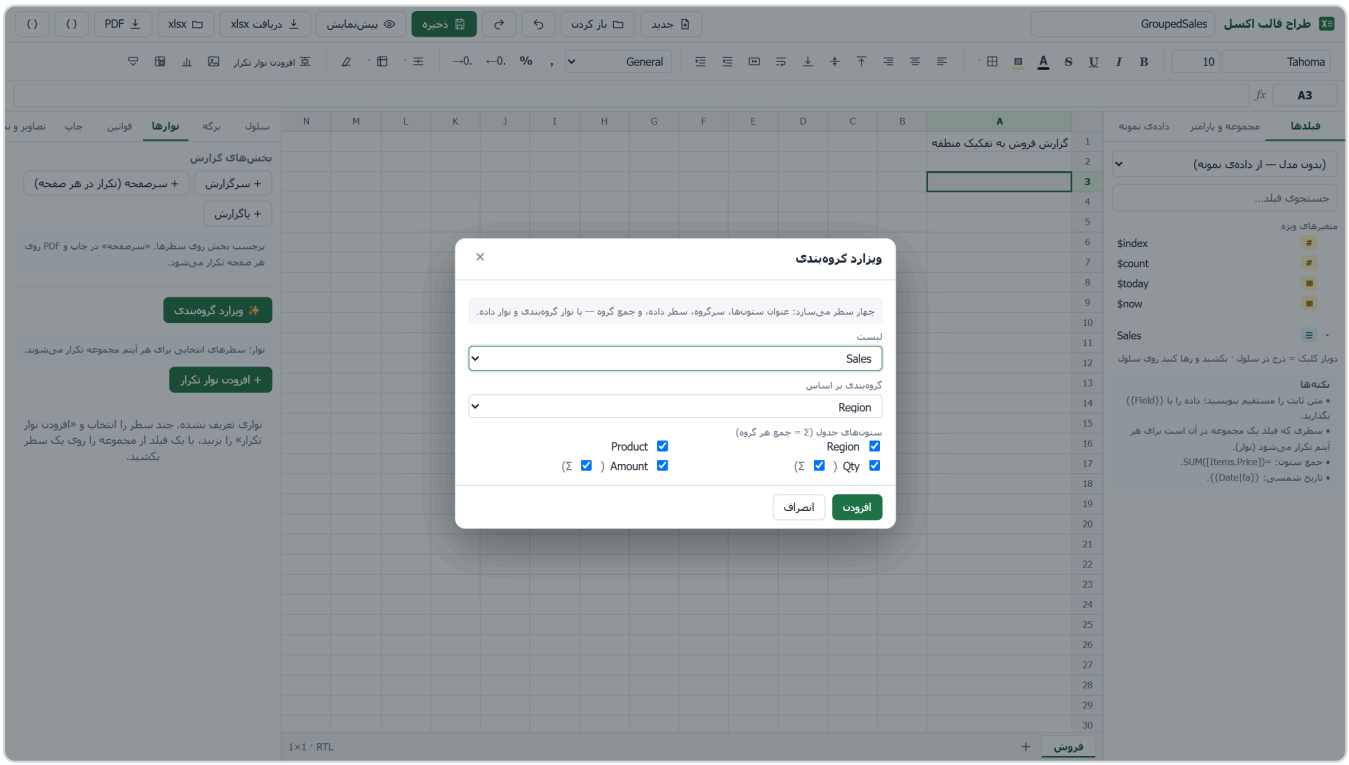
4.3 Fields & sample data

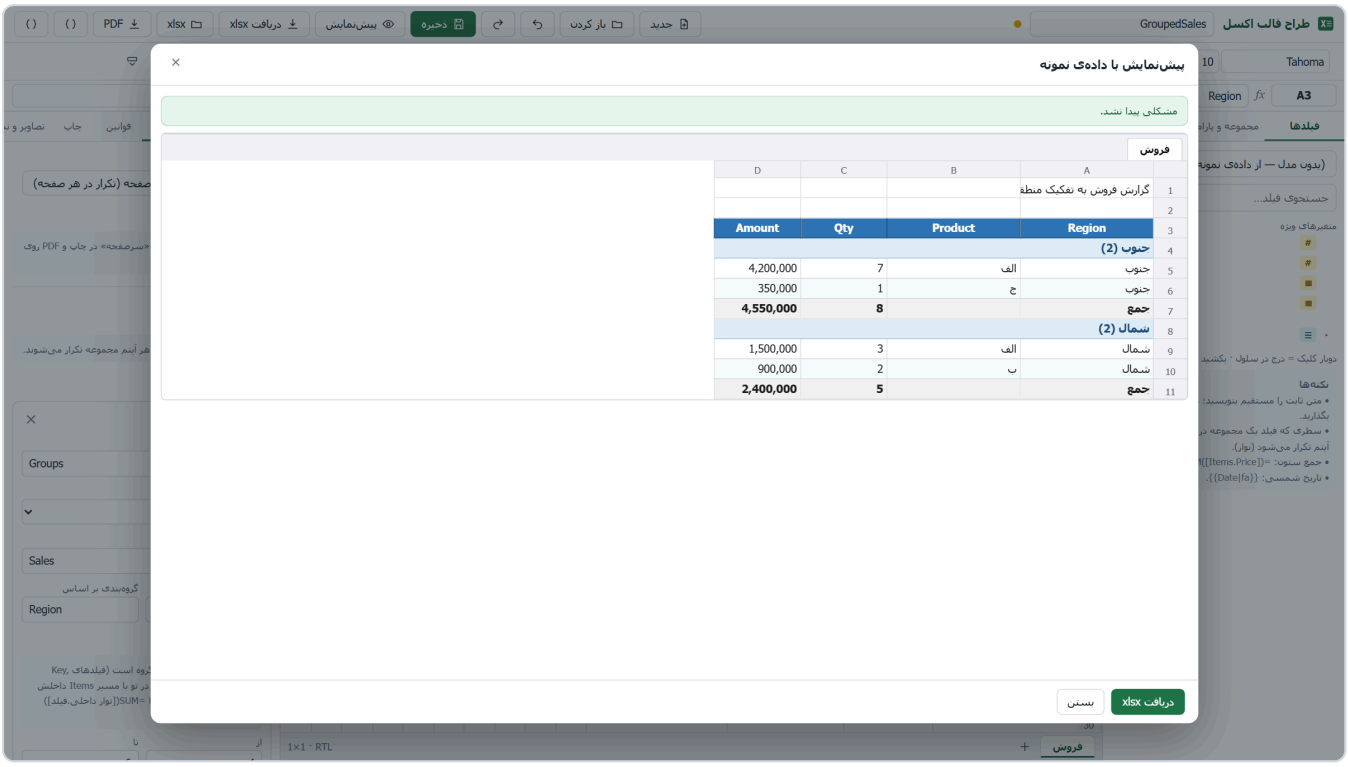
- Fields come from a **registered model** (`AddModel<T>`) or are inferred from the **sample JSON**.
- **Drag a field onto a cell** (or double-click it). If the field belongs to a list, **the repeat band for that row is created automatically**.
- Buttons on field nodes: **Σ** (sum of a numeric column) and **ش** (Persian/Jalali date).
- *Sample data* tab: sample JSON of the data you will pass (used only for the designer preview).

4.4 Bands & sections



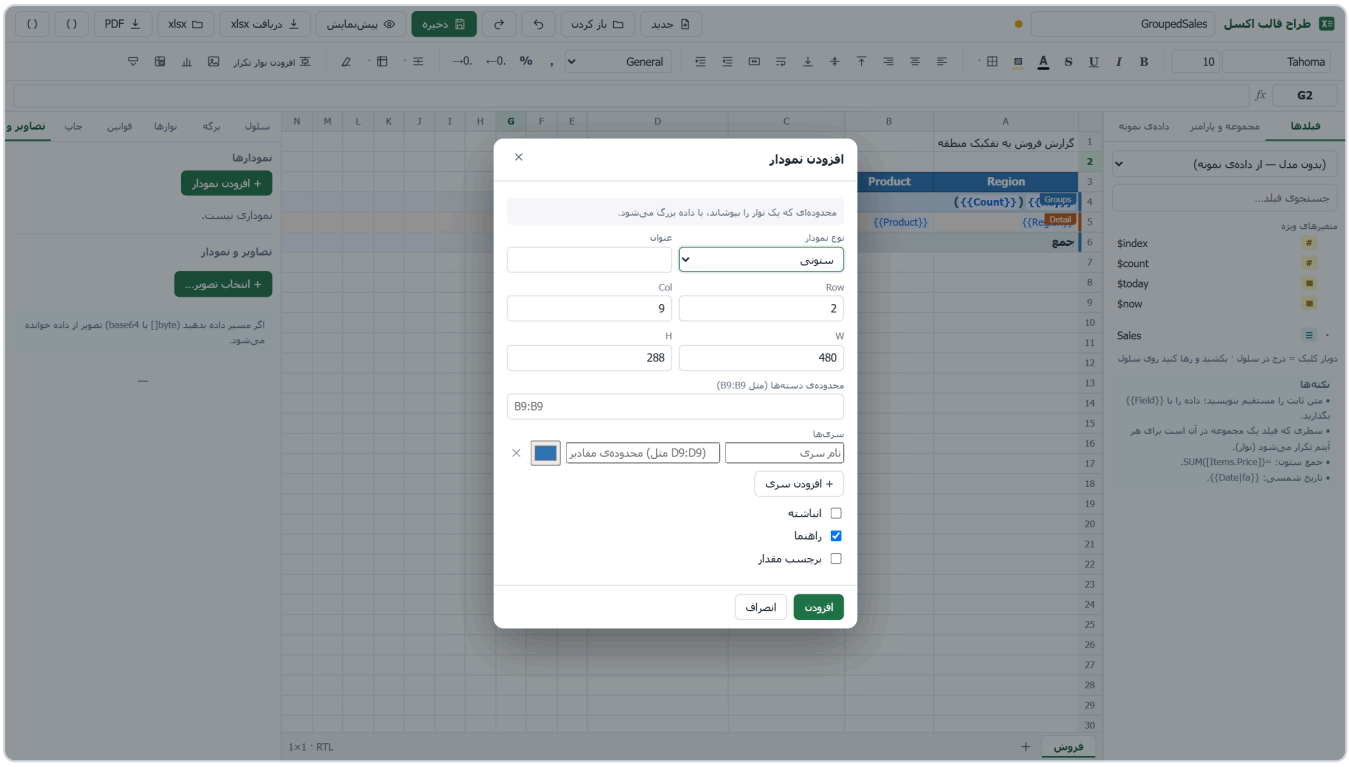
- **Repeat band:** collection path (`Lines`), name, rows, even-row colour, "keep a blank block when empty", **sort**, **group**, direction (rows/columns).
- **Sections:** report header · **page header** (repeats on every printed/PDF page) · report footer — coloured labels on the rows.
- **Group wizard ✨:** pick a list, the group field and the columns; it creates four rows (column titles, group header, detail, group total) plus the bands.





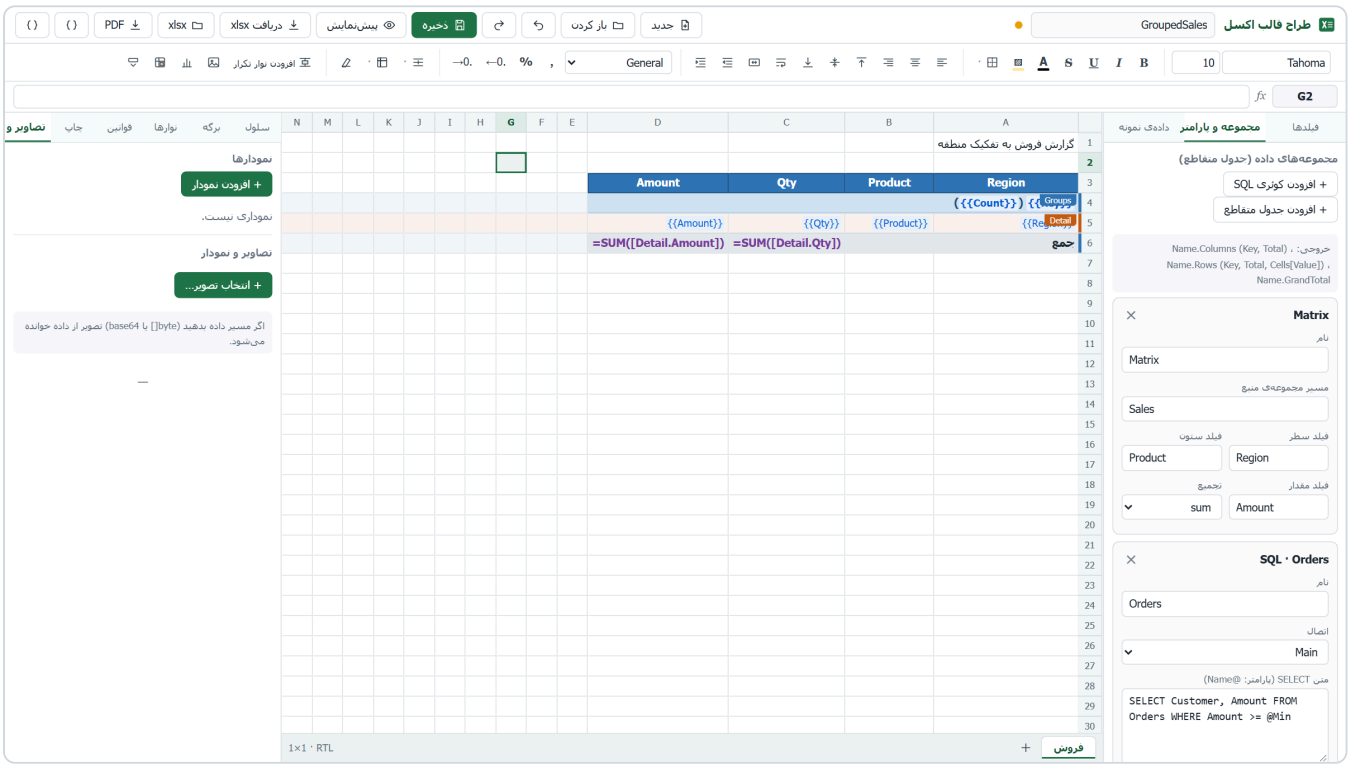
4.5 Other panels

- **Cell:** value/formula, value type (auto / always text), link, text direction.
- **Sheet:** name, right-to-left, gridlines, tab colour, zoom, default sizes, "repeat sheet per item", freeze, auto-filter, protect.
- **Rules:** conditional formats (cell value, formula, contains text, duplicates, top/bottom, colour scale, data bar) and validation (list, number, date, length, custom).
- **Print:** orientation, paper, fit to width, margins, header/footer (`&P &N &D &T &A &F`), repeat rows, print area.
- **Images & charts:** static or data-bound logo (`byte[]`/base64); charts with type, title, category range and series.

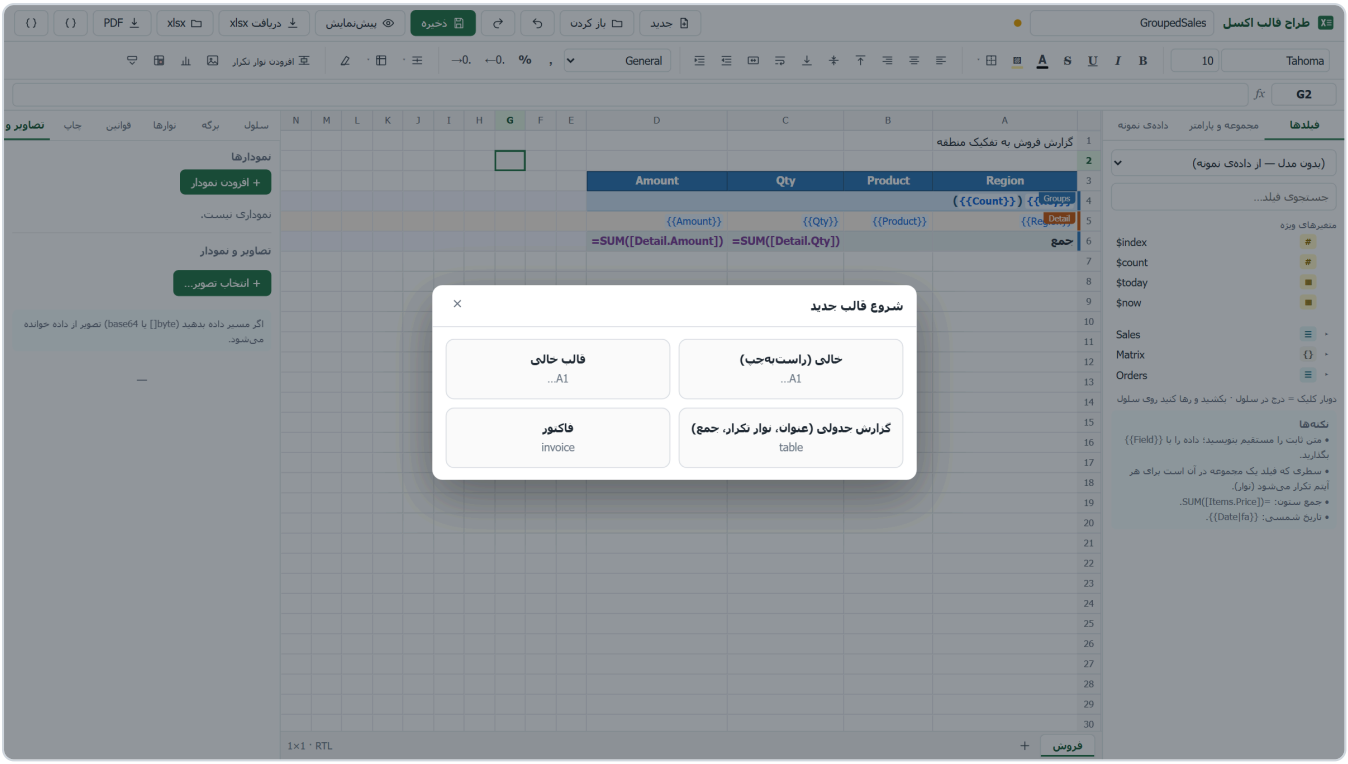


4.6 Data sets & parameters

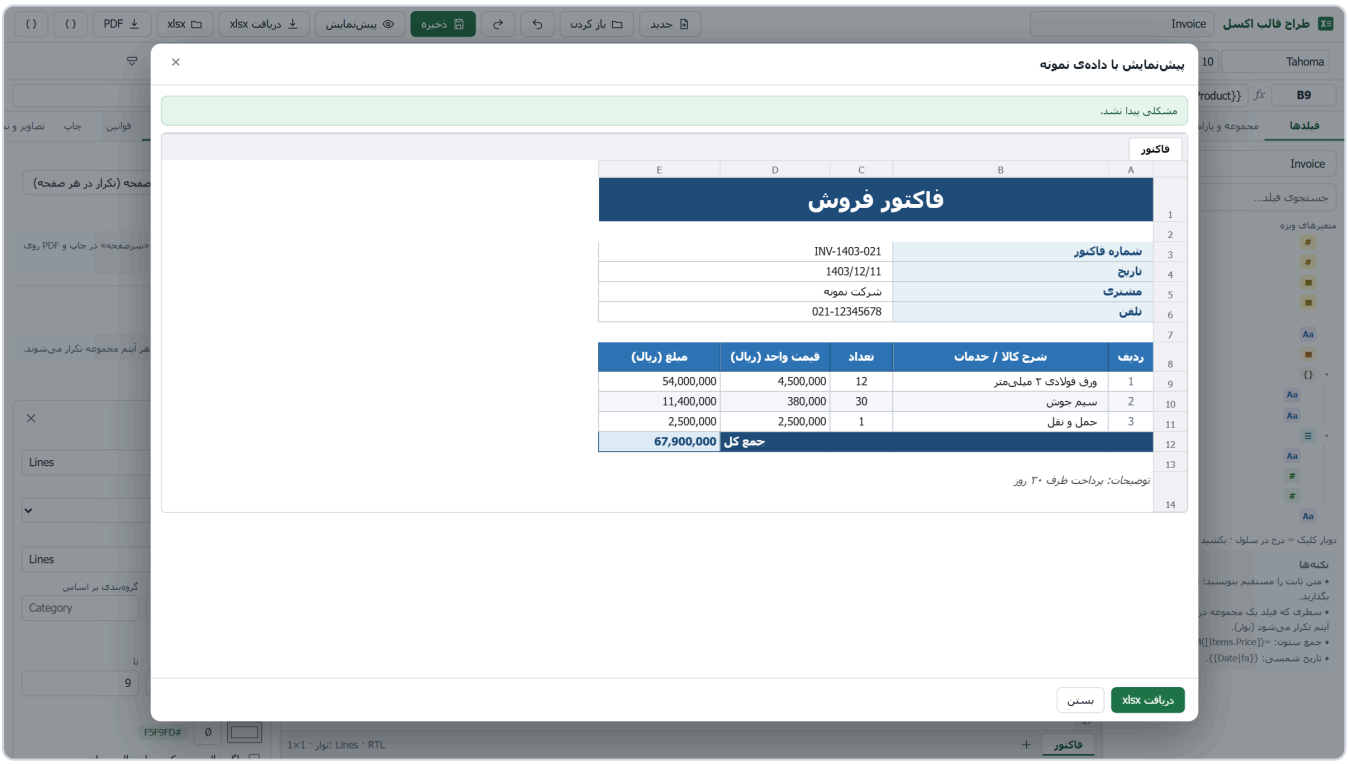
- **Cross-tab (pivot), SQL query, parameters.**



4.7 New template & preview



Preview runs the template on the sample data, shows every sheet as a tab and lists **template warnings**:



4.8 Autosave & moving templates

A draft of every change is kept in the browser and offered back after an accidental close. Real saving is **Save** to the server. **Export/Import JSON** moves templates between environments. `?t=Invoice` opens a template directly.

5. Tutorial: build an invoice

Goal: an invoice with customer details, a line-item table, a grand total, and Excel/PDF output (the result is the bundled `Invoice` sample).

1. Register the model (once, in `Program.cs`):

```
builder.Services.AddKscExcel(o => { o.AddModel<Invoice>("Invoice"); });
public class Invoice {
    public string Number { get; set; } = ""; public DateTime Date { get; set; }
    public Customer Customer { get; set; } = new(); public List<InvoiceLine> Lines { get; set; } =
new();
    public string Notes { get; set; } = "";
}
public class Customer { public string Name { get; set; } = ""; public string Phone { get; set; } = "";
}
public class InvoiceLine { public string Product { get; set; } = ""; public int Qty { get; set; }
public decimal UnitPrice { get; set; } }
```

2. Open the designer → **New** → "Blank (right-to-left)". In the Fields panel pick the `Invoice` model.

3. Title: select A1:E1 → **Merge** → type "Invoice" → font 20 bold, white text, dark-blue fill, centred.

4. Header block: drag `Number`, `Date`, `Customer.Name` onto cells. For a Persian date use the **ش** button on the `Date` node (`{{Date|fa}}`). Type labels next to them.

5. Table header: type the header row (#, Product, Qty, Unit price, Amount) and style it.

6. Detail row: in the next row put `{{$index}}`, then drag `Lines.Product`, `Lines.Qty`, `Lines.UnitPrice`. **The `Lines` band is created automatically** (coloured tag beside the row). Make the Amount column a formula: `=[Qty]*[UnitPrice]`.

7. Grand total: the row below the band: "Total" and `=SUM([Lines:E])` (column E = Amount). Apply number format `#,##0`.

8. Polish: Bands tab → enable the even-row colour; Sheet tab → turn gridlines off.

9. Preview → check warnings → **Save** as `Invoice`.

10. Use it:

```
public IActionResult Invoice(int id) => this.ExcelFile("Invoice", _repo.Get(id), $"invoice-
{id}.xlsx");
```

> Templates are plain files (`ExcelTemplates/Invoice.kxl.json`) — keep them in Git.

---

6. Template language

6.1 Text, numbers, values

- Fixed text is typed as is. A fixed number (`123`) is a number; `0912…` (leading zero) and strings over 15 digits stay **text** (Cell panel → "Always text").
- Data: `{{path}}`, e.g. `{{Customer.Name}}`, `{{Lines[0].Product}}`; case-insensitive.
- If the whole cell is **a single** `{{…}}`, the **type is kept** (number/date/bool); mixed with text it becomes a string.
- `null` = empty cell (use `|default:` for a fallback).

6.2 Special variables

Variable	Meaning
`{{$index}}`	1-based row number inside a band
`{{$count}}`	number of items in the band (unknown when streaming)
`{{$today}}` / `{{$now}}`	today / now (server)
`{{$root}}` / `{{$parent}}` / `{{$item}}`	root data / parent item / current item
`{{$p.Name}}`	report parameter

Inside a band, a name not found on the current item is looked up on the parent and then the root.

6.3 Filters `{{path | filter:arg}}`

Filter	Purpose	Example
`fa`	Persian (Jalali) date as text	`{{Date|fa}}` → `1404/01/01`, `{{Date|fa:yyyy/MM/dd HH:mm}}`
`fadigits`	Persian digits	`{{N|fadigits}}` → `۱۲۳۴`
`format:pattern`	.NET format	`{{Date|format:yyyy-MM-dd}}`
`round:n`	round	`{{P|round:1}}`
`upper` `lower` `trim`	case/whitespace	
`default:text`	fallback for empty	`{{X|default:-}}`
`yesno:Yes/No`	bool to text	`{{Active|yesno:Yes/No}}`
`join:,`	join a list	`{{Tags|join:, }}`

Filter	Purpose	Example
`truncate:n`	shorten with …	
custom	register with `AddFilter`	`{{Rating|stars}}`

> Without a filter a date becomes a **real (Gregorian) Excel date** with format `yyyy/mm/dd` (sortable); `|fa` yields Jalali **text**.

6.4 Aggregates & expressions

- Aggregates: `{{Count(Lines)}}`, `{{Sum(Lines.Qty)}}`, `Avg`, `Min`, `Max` (static values).
- Expressions: `{{Qty * UnitPrice}}` · `{{IF(Total > 1000, 'VIP', 'Normal')}}` · `{{ROUND(Price * 1.09, 0)}}` · `{{FirstName & ' ' & LastName}}` · `{{COALESCE(Nick, Name)}}` · `{{FA(Date, 'yyyy/MM/dd')}}`.
- Operators: `+ - * / % ^ & = <> < > <= >=`; functions: `IF IFERROR COALESCE ROUND ABS MIN MAX SUM AVG LEN UPPER LOWER TRIM CONCAT AND OR NOT TODAY NOW YEAR MONTH DAY FA`, plus your custom functions.

6.5 Repeat bands

Scenario	How
Simple list	a row with `{{Product}}` … and a band `Lines` on that row
Several rows per item	band over several rows
Nested	band `Groups` over rows 1–3 and band `Lines` over row 2 (relative path `Lines`)
Empty list	rows vanish; "keep a blank block when empty" keeps one
Zebra	even-row fill, except cells with an explicit fill
Sort / group	`SortBy`, `GroupBy` on a band; each group exposes `Key`, `Count`, `Items`

Grouping with group totals:

Row	Content	Band
1	`{{Key}} ({{Count}})`	`Groups` (path `Sales`, group by `Region`) over rows 1–3
2	`{{Product}}` · `{{Qty}}`	`Detail` (path `Items`) over row 2
3	`Total` · `=SUM([Detail.Qty])`	(part of `Groups`)

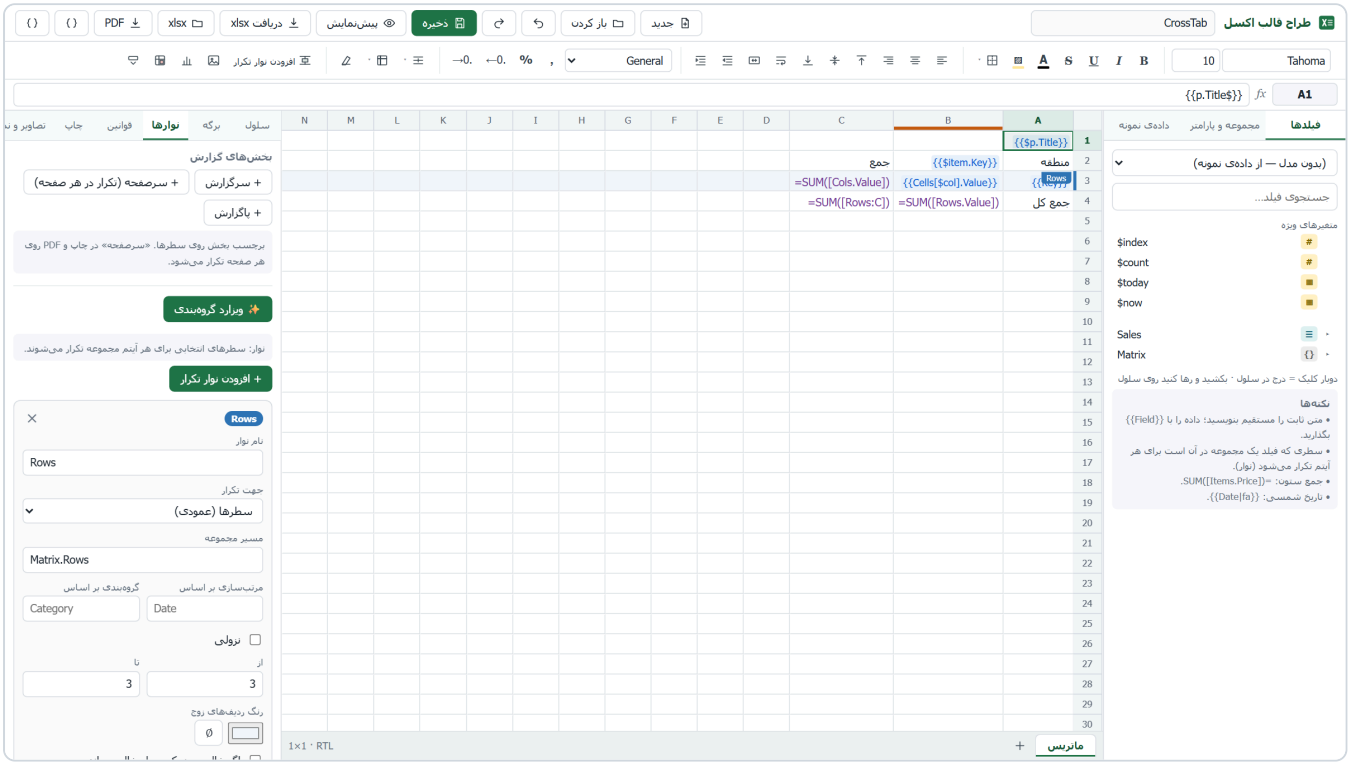
(The group wizard builds exactly this.)

6.6 Cross-tab

1. **Data sets & parameters** → "Add cross-tab": name `Matrix`, source `Sales`, row `Region`, column `Product`, value `Amount`, aggregate `sum`.

2. A **row** band `Rows` (path `Matrix.Rows`) over the data row.
3. A **column** band `Cols` (path `Matrix.Columns`, direction "columns", column 2 to 2).

Cell	Content
Column header (inside the column band)	`{{$item.Key}}`
Data (inside the column band)	`{{Cells[$col].Value}}`
Row total	`=SUM([Cols.Value])`
Column total	`=SUM([Rows.Value])` (inside the column band)
Grand total	`=SUM([Rows:C])`



In code: `Pivot.Build(data, "Region", "Product", "Amt", "sum")`.

6.7 Formulas & tokens

A cell starting with `=` is an Excel formula and stays live. Special tokens are resolved to real addresses at generation time:

Token	Meaning	Example
`[Field]`	the field's cell in the **same row**	`=[Qty]*[Price]` → `=C9*D9`
`[Band.Field]`	whole column of the field in the band	`=SUM([Lines.Qty])` → `=SUM($C$9:$C$11)`
`[Band:E]`	whole column `E` in the band	`=SUM([Lines:E])`

Token	Meaning	Example
`[ColBand.Field]`	**horizontal** range in the same row of a column band	`=SUM([Cols.Value])`

`{{…}}` inside a formula is substituted too (`=C9*{{TaxRate}}`). Empty band → token becomes `0`. **Cached values** are computed for common functions (`SUM AVERAGE MIN MAX COUNT COUNTA ABS ROUND IF IFERROR AND OR NOT CONCAT LEN UPPER LOWER TRIM SUMIF COUNTIF`); other Excel functions (e.g. `VLOOKUP`) are written as is and work in Excel. Fixed `A1` references are **not** adjusted when rows/columns move — use tokens.

6.8 Sheet repeat, links, dynamic images

- **Sheet repeat:** Sheet panel → "Repeat sheet per item" = `Branches` (the sheet name may contain `{{Name}}`).
- **Link:** Cell panel → link: `https://site/{{Id}}` or `#Sheet2!A1`.
- **Dynamic image:** data path of the image (`Company.Logo`, `byte[]` or base64).

6.9 Advanced reporting: multi-level groups, calculated columns, Jalali calendar

Multi-level grouping with subtotals. In the *Bands* tab, *Group wizard* now offers a **second grouping level**, a **grand total row** and group keys written as **expressions**. It builds nested bands (`Groups` → `SubGroups` → `Detail`) with a header and a subtotal row per level. Totals use `SUBTOTAL(9, …)`, exactly like Excel's *Subtotal* command: a total over a range ignores the subtotals inside it, so nested totals never double count (the engine computes the cached values the same way). By hand: put the same formula `=SUBTOTAL(9, [Detail.Amount])` in every subtotal row; a token such as `[Detail.Amount]` now covers **all** detail rows below the current group, including those of nested groups.

Group / sort by an expression. `GroupBy` and `SortBy` accept an expression when they start with `=`: `=JYEAR(Date)`, `=JYM(Date)` (Jalali year/month), `=IF(Qty>10,"big","small")`. In the designer type it in the wizard's "or expression" box.

Calculated columns (inside `{{ }}`, evaluated per band item over the whole band, or per group inside a group):

Function	Meaning
`RUNSUM(Qty)`	running total up to this row
`PCT(Qty)`	share of the band total (0-1; format the cell as `0%`)
`PREV(Qty)` / `DIFF(Qty)`	previous row's value / difference from it
`RANK(Qty)`	rank, 1 = largest
`MOVAVG(Qty, 3)`	moving average of the last 3 rows

They need the whole collection, so they are not available in streaming mode (the cell is empty there).

Jalali (Persian) calendar. Dates stay real Gregorian dates in the file; these helpers do the Jalali part:

Expression / filter	Result
`JYEAR(D)` `JMONTH(D)` `JDAY(D)` `JQUARTER(D)`	numbers
`JMONTHNAME(D)`, `{{D|jmonthname}}`	month name (فروردین …)
`JYM(D)`, `{{D|jym}}`	`1404/02` text, handy as a group key
`JDAYNAME(D)`	weekday name
`JDATE(1404,1,1)`	Jalali date → real date
`JPARSE('1404/02/10')`	text (Persian digits allowed) → real date
`{{D|fa}}`	`1404/02/06` text (as before)

Report parameters of type `jdate` accept Jalali text from the designer default or from code: `new Dictionary<string, object> { ["From"] = "1404/01/01" }`; the template sees a real date (`{{$p.From|fa}}`, SQL `@From`).

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7. Developer guide — .NET 8/10

7.1 Setup

```
using KSC.Excel.AspNetCore;
using KSC.Excel.Pdf;

builder.Services.AddControllersWithViews();
builder.Services.AddKscExcel(o =>
{
    o.AddModel<Invoice>("Invoice");      // fields + sample data for the designer
    o.UsePdf();                          // PDF button in the designer (optional)
});

var app = builder.Build();
app.UseRouting();
app.UseAuthentication(); app.UseAuthorization();

app.MapKscExcelDesigner("/excel-designer").RequireAuthorization("Admin");  // the designer
app.MapKscExcelDownloads();                                                 // viewer download links
app.MapDefaultControllerRoute();
```

Templates live in `{ContentRoot}/ExcelTemplates/*.kxl.json` by default. Ship them: `<Content Include="ExcelTemplates\**" CopyToOutputDirectory="PreserveNewest" />`.

7.2 Producing files

```
public class ReportsController : Controller
{
    private readonly IExcelReportService _excel;
    public ReportsController(IExcelReportService excel) => _excel = excel;

    public IActionResult Invoice(int id) =>
        this.ExcelFile("Invoice", _repo.Get(id), $"invoice-{id}.xlsx");                 // download

    public IActionResult Sales(DateTime from) =>
        this.ExcelFile("Sales", _repo.Query(from), new Dictionary<string, object> { ["From"] = from },
"sales.xlsx");   // with parameters

    public IActionResult Big() => this.ExcelFileStreaming("Big", _repo.StreamRows(), "big.xlsx");   //
streaming

    public IActionResult InvoicePdf(int id) =>
        File(_excel.RenderPdf("Invoice", _repo.Get(id)), "application/pdf", "invoice.pdf");        //
PDF
}
```

Minimal API: `app.MapGet("/r", (IExcelReportService x) => x.File("Invoice", data, "invoice.xlsx"));`

Without a web host (job/service):

```
byte[] bytes = svc.Render("Invoice", model);
File.WriteAllBytes("invoice.xlsx", bytes);
svc.Render(svc.GetTemplate("Invoice"), model, outputStream);     // straight to a Stream
```

7.3 Showing the report in a page (viewer)

```
@await Component.InvokeAsync("KscExcelViewer", new { template = "Invoice", data = Model, downloadName
= "invoice" })
@* or: @addTagHelper *, KSC.Excel.AspNetCore  →  <vc:ksc-excel-viewer template="Invoice" data="Model"
/> *@
```

Parameter	Default	Description
`template`	—	template name
`data`	—	data
`downloadName`	template name	download file name
`showDownload`	`true`	download button
`showHeaders`	`false`	Excel-like A,B,C / 1,2,3 headers

Parameter	Default	Description
`downloadLabel`	"Download Excel" (fa: Persian)	button text

The download link is an unguessable 144-bit token, valid 15 minutes, bound to the signed-in user, served from the memory of the same server (max 64 files).

7.4 Input data

`data` can be a class/DTO, `Dictionary<string, object>`, `ExpandoObject`, `JObject/JToken`, `DataTable/DataRow/DataView`, or a mix. Names are case-insensitive. For streaming pass **lazy** sequences (`IEnumerable` with `yield`, a `DataReader`).

7.5 `IExcelReportService`

Method	Purpose
`Render(name, data)` / `(name, data, parameters)`	`byte[]` xlsx from a stored template
`Render(template, data)` / `(template, data, Stream)`	from a template object / into a Stream
`RenderStreaming(name|template, data, Stream, parameters?)`	streaming; returns `bool` (false = fell back to normal render)
`RenderHtml(name|template, data, showHeaders)`	HTML for a page
`Bind(template, data, parameters?)`	`RenderedWorkbook` + `Warnings`
`BindPreview(template, data, parameters?)`	like `Bind`, but skips SQL when not allowed (designer)
`GetTemplate(name)` / `Store`	template / storage
`svc.RenderPdf(...)` (Pdf package)	PDF

7.6 Code-first templates

```
var b = new TemplateBuilder("Report", "Report");
var head = b.Style("h", new StyleDef { Bold = true, Fill = "#2E75B6", Color = "#FFFFFF", H = "center"
});
b.Sheet("Report", s =>
{
    s.Cell("A1", "Name", head).Cell("B1", "Amount", head);
    s.Cell("A2", "{{Name}}").Cell("B2", "{{Amount}}");
    s.Band("Rows", "Rows", 2, 2, "#F5F9FD");
    s.Cell("A3", "Total").Cell("B3", "=SUM([Rows.Amount])");
    s.Freeze(1);
});
byte[] xlsx = svc.Render(b.Build(), new { Rows = data });
```

Good for default templates shipped with your app (`store.Save(template)` at startup).

7.7 Template storage (`ITemplateStore`)

```
public sealed class DbTemplateStore : ITemplateStore
{
    public IList<TemplateInfo> List() => /* SELECT Name, Title, Modified */;
    public ExcelTemplate Get(string name) { var json = /* SELECT Json */; return json == null ? null :
ExcelTemplate.FromJson(json); }
    public void Save(ExcelTemplate t) { /* UPSERT t.Name, t.ToJson() */ }
    public bool Delete(string name) => /* DELETE */;
}
builder.Services.AddKscExcel(o => o.Store = new DbTemplateStore(...));
```

Validate names with `TemplateNames.IsValid(name)`.

7.8 Testing templates

```
var wb = svc.Bind(svc.GetTemplate("Invoice"), sampleData);
Assert.Empty(wb.Warnings);                      // a warning-free template
var bytes = XlsxWriter.ToBytes(wb);             // inspect with ZipArchive if needed
```

---

8. Developer guide — .NET Framework (MVC 5)

Reference `KSC.Excel` and `KSC.Excel.Mvc` (adapter targets `net472` / `net48`; the core works from 4.6.1).

```
// Global.asax.cs
protected void Application_Start()
{
    AreaRegistration.RegisterAllAreas();
    KscExcel.Initialize(o => { o.TemplateDirectory = "ExcelTemplates"; o.AddModel<InvoiceVm>
("Invoice"); });
    KscExcel.Authorize = ctx => ctx.User.IsInRole("Admin");      // optional; the user must be signed
in
    RouteConfig.RegisterRoutes(RouteTable.Routes);
}

// RouteConfig.cs  (before the default MVC route)
routes.MapKscExcelDesigner();          // /excel-designer
routes.MapKscExcelDownloads();         // for Html.KscExcelViewer (after Initialize)
```

```
// Controller
using KSC.Excel.Mvc;
public ActionResult Invoice(int id) => this.Excel("Invoice", _svc.Get(id), "invoice.xlsx");
public ActionResult Sales() => this.Excel("Sales", data, new Dictionary<string, object> { ["From"] =
DateTime.Today }, "sales.xlsx");
public ActionResult Big() => this.ExcelStreaming("Big", reader, "big.xlsx");
```

```
@using KSC.Excel.Mvc
@Html.KscExcelViewer("Invoice", Model, "invoice")
```

- Without DI use `KscExcel.Service` (the same `IExcelReportService`).
- IIS: Integrated pipeline. If `/excel-designer/assets/*.js` returns 404, check `UrlRoutingModule` or set `<modules runAllManagedModulesForAllRequests="true" />`.
- **Verified on real IIS Express** (.NET Framework 4.8, System.Web + MVC 5.2.9): the sample `samples/KSC.Excel.Sample.Mvc5` serves the designer, viewer, PDF viewer, Excel and PDF downloads. Run it: `iisexpress.exe /path:samples\KSC.Excel.Sample.Mvc5 /port:5290`.
- **PDF works on .NET Framework** (`KSC.Excel.Pdf` is `netstandard2.0`): call `o.UsePdf()` in `KscExcel.Initialize`, then `this.Pdf("Invoice", model, "invoice.pdf")` or `Html.KscPdfViewer(...)`. SkiaSharp needs binding redirects in `web.config` (System.Runtime.CompilerServices.Unsafe 6.0.3.0, System.Memory, System.Buffers, System.Numerics.Vectors) - the sample's `web.config` has them; classic MVC projects get them automatically from *Add binding redirects*.

---

9. Windows desktop designer

The same web designer runs in `WebView2` (**.NET 10 or .NET Framework 4.8** - pick the build that matches the machine) and its **API is served in-process** (`https://kxl.local`; no Kestrel, no port). Look, features and the **template format are identical**.

```
dotnet publish src/KSC.Excel.Designer.Windows -c Release -r win-x64 --self-contained false -o
artifacts/KSC.Excel.Designer
artifacts\KSC.Excel.Designer\KSC.Excel.Designer.exe            # or:  KSC.Excel.Designer.exe
"C:\path\Invoice.kxl.json"
```

- Templates default to `Documents\KSC Excel Templates`; **File → Choose templates folder** lets you point at your project's folder (e.g. `ExcelTemplates` in Git). The setting is stored in `%APPDATA%\KSC.Excel.Designer\settings.json`.
- Menus: New · **Open `.kxl.json` file** · Save · Save as JSON · **Import from Excel file** · **Export Excel and PDF with sample data** · choose/show folder · reload · DevTools.
- Limitation: `AddModel<T>` and SQL connections cannot be registered in the desktop app; provide sample JSON instead.

- Self test: `KSC.Excel.Designer.exe --selftest out.txt`.

---

10. PDF export

```
using KSC.Excel.Pdf;
byte[] pdf = svc.RenderPdf("Invoice", model);                          // with parameters:
svc.RenderPdf("Sales", data, parameters)
ExcelPdf.Render(svc.Bind(template, data, null), stream, new PdfOptions { FontFile =
"/fonts/Vazirmatn.ttf" });
byte[] png = ExcelPdf.RenderPng(wb, pageNumber: 1);                    // image preview
```

فاکتور فروش

شماره فاکتور	INV-1403-021
تاریخ	1403/12/11
مشتری	شرکت نمونه
تلفن	021-12345678

ردیف	شرح کالا / خدمات	تعداد	قیمت واحد (ریال)	مبلغ (ریال)
1	ورق فولادی ۲ میلی‌متر	12	4,500,000	54,000,000
2	سیم جوش	30	380,000	11,400,000
3	حمل و نقل	1	2,500,000	2,500,000
	جمع کل			67,900,000

توضیحات: پرداخت ظرف ۳۰ روز

	Supported
Persian/Arabic text	correct shaping (HarfBuzz), mixed Persian + digits + Latin ordering (bidi), RTL sheets
Cells	fill, borders, font/size/bold/italic/underline, alignment, wrap, indent, merges, number formats, formulas
Pages	A4/A3/A5/Letter/Legal, orientation, margins, scale/fit-to-width, **repeating title rows and page header**, header/footer (`&P &N &D &T &A &F &L &C &R`), pagination
Images/charts	images; column, bar, line, area, pie charts
Links	clickable `http/https/mailto`

Not supported: auto-filter arrows, cell comments. The result is very close to, but not pixel-identical with, Excel's own "Save as PDF".

Fonts: cell font → `FallbackFontFamily` (Tahoma) → Segoe UI/Arial/Noto/DejaVu → OS font manager. On **Linux/Docker** install a Persian font or set `PdfOptions.FontFile`, and add `SkiaSharp.NativeAssets.Linux.NoDependencies`. Page cap: `PdfOptions.MaxPages` (2000).

10.1 PDF viewer in the page

Show the report as a PDF inside your page (the browser's PDF reader: page navigation, zoom, search, print) with **Download PDF / Download Excel / Print** buttons. No JavaScript library, no CDN.

```
@* ASP.NET Core *@
@await Component.InvokeAsync("KscPdfViewer", new { template = "Invoice", data = Model, downloadName =
"invoice", showExcel = true, height = "75vh" })

@* .NET Framework MVC 5 *@
@Html.KscPdfViewer("Invoice", Model, "invoice", height: "75vh", showExcel: true)
```

Requirements: `options.UsePdf()` (KSC.Excel.Pdf) and the download route (`app.MapKscExcelDownloads()` / `routes.MapKscExcelDownloads()`). Controller one-liners:

```
return this.PdfFile("Invoice", model, "invoice.pdf");                       // ASP.NET Core  (inline:
true = show in the browser)
return this.Pdf("Invoice", model, "invoice.pdf");                           // MVC 5         (inline:
true = show in the browser)
app.MapGet("/r", (IExcelReportService s, KscExcelOptions o) => s.Pdf(o, "Invoice", model));   //
minimal API
```

Parameters of the viewer: `template`, `data`, `parameters` (report parameters), `downloadName`, `height` (CSS), `showToolbar`, `showExcel`.

10.2 What PDF supports

Everything in the table above **plus**: conditional formatting (same engine as the HTML preview), rotated and vertical text, text that spills into empty neighbouring cells (like Excel), shrink-to-fit, data bars / colour scales, **barcodes and QR codes as crisp vector shapes**, **watermark**, **page break per group**, and crisp vector output for everything except images.

10.3 Barcodes and QR codes

Select a cell, set **Barcode** in the Cell panel (`QR Code`, `QR Code (L)`, `QR Code (H)`, `Code 128`) and write the content as usual: `{{Number}}`, `https://pay.example.com/{{Id}}`. The text is encoded and drawn as an image that fills the cell (merge cells for a bigger code). It works inside repeating bands (one code per row), in `.xlsx` (PNG image), in the HTML preview as a marker and in **PDF as vector shapes**.

- **QR:** byte mode (UTF-8, so Persian works), versions 1-10 (about 270 bytes at level L), ECC L/M/Q/H (`qr:L`, `qr:M`, `qr:Q`, `qr:H`; default M). Verified against an independent decoder in the test-suite.
- **Code 128:** printable ASCII; digit-only values of even length use the compact set C automatically.
- Text that cannot be encoded produces a warning and no image (the rest of the report still renders).
- Template JSON: `{ "r": 14, "c": 1, "v": "{{Number}}", "bc": "qr" }`. Barcode cells are not available for the streaming writer.

10.4 Watermark and page breaks

- **Watermark** (PDF): Print panel → *Watermark*, or `PdfOptions.Watermark = "DRAFT"` (overrides the template). Diagonal, light grey, on every page, any script.
- **Page break per item**: Bands panel → *Start each item on a new page* (`BandDef.PageBreak`) - one invoice / one group per page. Written to `.xlsx` as manual page breaks (`rowBreaks`) and honoured by the PDF paginator.

10.5 Password protection and permissions

```
var pdf = svc.RenderPdf("Invoice", model, null, new PdfOptions
{
    Security = new PdfSecurity { UserPassword = "1234", OwnerPassword = "boss", AllowCopy = false,
AllowModify = false }
});
```

AES-256 (PDF 2.0 standard security handler). `UserPassword` is needed to open the file; with an empty user password anyone can open it but the denied permissions (`AllowPrint`, `AllowCopy`, `AllowModify`, `AllowAnnotate`) are enforced by compliant readers, and the owner password lifts them. Verified by decrypting with an independent library. Note that permissions are advisory for readers that choose to ignore them; use a user password when confidentiality matters.

---

11. Huge reports: streaming

```
bool streamed = svc.RenderStreaming("BigReport", new { Title = "…", Items = repo.StreamRows() },
outputStream);
```

- Measured: **1,000,000 rows × 6 columns ≈ 12.6 s, ≈ 47 MB peak memory** (30 MB file); 100 k rows ≈ 2 s. Only `MaxRows` applies.
- If a template is not streamable, `RenderStreaming` falls back to the normal render and returns `false`; reasons: `StreamAnalyzer.Check(template)`. Rules: totals must be **below** their band, `[Field]` must not refer to a later row, no `{{$count}}`, frozen rows above every band.
- Differences: strings are `inlineStr` and formulas have **no cached values** (Excel calculates on open).
- An error mid-write leaves a partial file in `output`; the adapters write to a temp file.

---

12. SQL data sets, parameters, custom functions

Parameters

Data sets & parameters tab → name, type (`text/number/date/bool`), default; in the template `{{$p.Title}}`; in code:

```
svc.Render("Sales", data, new Dictionary<string, object> { ["Title"] = "Spring sales", ["From"] =
DateTime.Today });
```

SQL

```
o.AllowSql = true;                                                    // default false
o.AddConnection("Sales", SqlClientFactory.Instance, connectionString);       // any DbProviderFactory
o.SqlCommandTimeoutSeconds = 30; o.SqlMaxRows = 100000;
```

In the designer: "Add SQL query" → name (`Orders`), connection, `SELECT … WHERE Amount >= @Min` (`@Min` is bound from the report parameter with that name — **never string concatenation**). "Test query" discovers the columns. Then a band with path `Orders`. **Safety:** a single `SELECT`/`WITH` only; `INSERT/UPDATE/DELETE/DROP/EXEC/INTO/;` are rejected. This is only the second layer — **use a read-only database login**. With `AllowSql=false` and a SQL template, generation fails with a clear error; the designer preview uses same-named sample data instead.

REST / JSON / code providers (any data source)

SQL is one source; the engine can read **anything** through three more kinds of data sets. The template only *names* a source; **what the source is, is decided in your code** (a template author can never point the

engine at another server).

```
builder.Services.AddKscExcel(o =>
{
    // 1) your own code returns the data (EF, Dapper, a service call...) from the report parameters
    o.AddProvider("Orders", p => db.Orders.Where(x => x.Year == (int)p["Year"]).ToList());

    // 2) a REST endpoint: fixed host + headers (API key); templates give only the path
    o.AddHttpEndpoint("Crm", "https://crm.example.com/api/", new Dictionary<string, string> { ["X-Api-Key"] = key }, timeoutSeconds: 30);
});
```

In the designer (*Data sets & parameters* tab): **+ Add JSON / REST data** or **+ Add code provider**.

Data set type	Fields	Example
`provider`	`connection` = provider name	`Orders`
`json` + endpoint	`connection` = endpoint, `url` = path/query (`{Name}` = URL-encoded report parameter), `jsonPath` = where the array is	`orders?year={Year}`, `data.items`
`json` inline	`json` = fixed JSON text (lookup tables, constants)	`[{"Code":"A","Title":"Alpha"}]`

Then use the data set name as a band source (`Orders`, `Orders.Lines`...) or as a cross-tab source, like a SQL data set. **Test** loads the data set, shows the row count and fills the field tree with the discovered columns.

Safety: only http/https; a URL cannot leave the registered host/scheme; 30 s timeout and 20 MB response cap by default (`HttpEndpoint`); in the designer preview a failing source becomes a warning and the sample data is used.

Custom filters & functions

```
o.AddFunction("Tax", a => Convert.ToDecimal(a[0]) * 0.09m);      // {{Tax(Price)}} and the cached value of =TAX(B2)
o.AddFilter("stars", (value, arg) => new string('*', Convert.ToInt32(value)));   // {{Rating|stars}}
```

---

13. Import from an Excel file

The **xlsx** button (or the desktop menu) converts an `.xlsx` into a template: sheets, values, **formulas**, fonts/colours (including theme colours)/borders/alignment/number formats, merges, widths/heights, freeze panes, filter, RTL, conditional formats, validation, hyperlinks, **images**, print setup. Then replace data cells with `{{Field}}` placeholders and add bands. **Not imported:** charts, PivotTables, Excel tables (come in as

plain cells), macros. Safety: no XXE, 50 MB / 500 k cell limits. In code: `XlsxImporter.Import(bytes, "Name").Template`.

---

14. Reference: options & API

`KscExcelOptions`

Option	Default	Description
`TemplateDirectory`	`ExcelTemplates`	template folder (relative to ContentRoot / site root)
`Store`	files	custom storage
`DesignerTitle`	`KSC Excel Designer`	title
`Language`	`fa`	`fa` (RTL) or `en`
`ReadOnlyDesigner`	`false`	no save/delete
`AllowAnonymousDesigner`	`false`	otherwise sign-in is required outside Development
`DownloadPath`	`/kxl-download`	viewer link prefix
`MaxRows` / `MaxCells`	1,048,576 / 3,000,000	caps (`MaxCells` not applied when streaming)
`PreviewMaxRows`	500	preview rows
`AllowSql`, `AddConnection`, `SqlCommandTimeoutSeconds`, `SqlMaxRows`	`false`, —, 30, 100,000	SQL
`AddProvider(name, fn)` / `AddHttpEndpoint(name, baseUrl, headers)`	—	register code providers / REST endpoints for `provider` / `json` data sets
`AddFilter` / `AddFunction`	—	customisation
`AddModel<T>(name)` / `AddModelFromJson`	—	models for the designer
`UsePdf()` (Pdf package)	—	PDF button in the designer

Designer HTTP API

Prefix `{base}/api`; writes require the header `X-KXL-Request: 1`.

Route	Purpose
`GET /config`	title, language, models, starters, `pdf`, `allowSql`, `connections`
`GET /templates` · `GET/PUT/DELETE /templates/{name}`	list and CRUD
`GET /models/{name}` · `GET /starters/{key}`	model / starter template
`POST /preview` · `POST /download` · `POST /pdf` · `POST /validate`	preview, xlsx, PDF, warnings
`POST /import`	import xlsx (`{name, data(base64)}`)
`POST /sql/test`	test a query (only with `AllowSql`)

Template JSON

`schemaVersion`, `sheets[].cells[]`, `styles{}`, `bands[]`, `dataSets[]`, `parameters[]`, … Full schema: ../KSC.Excel-Design.md §5 (Persian). Templates diff well and belong in Git.

---

15. Security

- **The designer can write templates.** Use `RequireAuthorization("policy")` (Core) or `KscExcel.Authorize` (Framework). Outside Development anonymous users are rejected unless `AllowAnonymousDesigner`.
- In production usually set `ReadOnlyDesigner = true` and deploy templates from Git.
- Writes need a custom header (CSRF protection); template names allow only letters/digits/space/ `_ - .` ; `..` is rejected.
- User data never becomes a **formula** (no formula injection); preview HTML is encoded; links only `http/https/mailto`.
- xlsx import: no XXE, size/cell limits. SQL: off by default, SELECT only, parameterised, read-only login.
- Designer request cap 16 MB; row/cell/page caps.
- A template holds structure and `sampleData` only — do **not** store real/sensitive data in it.

---

16. Deployment

1. Copy the `ExcelTemplates` folder to the output.
2. Web farm: keep templates in a shared folder or database (`ITemplateStore`); viewer links live in the memory of one server (sticky sessions or direct download).
3. Docker: mount the templates folder as a volume; install a Persian font for PDF.

4. Fonts: the xlsx does not embed fonts; the template font (e.g. `B Nazanin`) must exist on the user's machine (default Tahoma).

---

17. Performance & limits

Rows × 6 cols	Time	File	Extra memory
1,000	≈ 40 ms	37 KB	≈ 1 MB
10,000	≈ 0.6 s	337 KB	≈ 15 MB
100,000	≈ 2.8 s	3.3 MB	≈ 80 MB
1,000,000 (streaming)	≈ 12.6 s	30 MB	≈ 47 MB

- The normal render builds the whole report in memory (`MaxCells`); use streaming for big ones.
- Excel limits: 1,048,576 rows, 16,384 columns, sheet names ≤ 31 characters.
- Concurrent renders are safe.

---

18. Troubleshooting & FAQ

Problem	Cause / fix
Designer returns 401	Sign-in required outside Development (or `AllowAnonymousDesigner`)
Save returns 403	`ReadOnlyDesigner` is on
Cell is empty	check the field path in the preview; `null` = empty (`|default:`)
`#REF!` / "unknown band"	the band name in `[Band.Field]` must match exactly and a cell in that band must contain `{{Field}}`; for a formula column use `[Band:E]`. Warnings appear in the preview and `Warnings`
Column total is 0	the band is empty (intentionally no error)
`The report has more than … rows/cells`	filter the data, use streaming, or raise `MaxCells`
Template not found after publish	set `ExcelTemplates` to `CopyToOutputDirectory`
Persian shows boxes in PDF (Linux)	install a Persian font / `FontFile`
Columns not right-to-left	Sheet panel → Right-to-left

Problem	Cause / fix
SQL does not run	`AllowSql`, `AddConnection`, connection name, read-only login
Edit the UI without rebuilding	env var `KSC_EXCEL_UI_DIR` = `src/KSC.Excel/Designer/UI` (tool development only)
Is Excel needed on the server?	No — no COM/Interop
Old `.xls`?	No — `.xlsx` only

---

19. Building from source, tests, contributing

```
src/KSC.Excel/                  core (Model, Engine, Writer, Import, Preview, Services, Designer +
UI)
src/KSC.Excel.AspNetCore/       ASP.NET Core adapter
src/KSC.Excel.Mvc/              .NET Framework adapter
src/KSC.Excel.Pdf/              PDF (SkiaSharp + HarfBuzz)
src/KSC.Excel.Designer.Windows/ desktop app (WPF + WebView2)
samples/KSC.Excel.Sample.Web/   runnable sample
tests/KSC.Excel.Tests/          tests (net10.0 and net48)
tests/KSC.Excel.Pdf.Tests/      PDF tests
docs/                           documentation
```

```
dotnet build
dotnet test tests/KSC.Excel.Tests          # net10.0 and net48
dotnet test tests/KSC.Excel.Pdf.Tests
```

Tests validate the xlsx package structure with `XDocument`; final validity was checked by opening files in **real Excel** (COM). UI without rebuilding: `KSC_EXCEL_UI_DIR`. When contributing: behaviour change → test; incompatible template JSON change → bump `schemaVersion`; update the docs.

---

20. Roadmap & known limitations

Done: designer, engine, bands/groups/cross-tab, charts, parameters, SQL, **REST/JSON/provider data**, expressions & custom functions, streaming, PDF (**.NET Framework + .NET**, conditional formats, rotation, overflow), **PDF viewer**, **barcodes/QR**, **watermark**, **page break per group**, xlsx import, desktop app (.NET 10 + .NET Framework 4.8), group wizard, report sections, real-IIS verification of the MVC 5 adapter.

Limitations: 1. Fonts are not embedded (the font must exist on the user's machine). 2. Memory in the normal render (use streaming). 3. HTML preview and PDF approximate Excel (images/barcodes are not drawn in the HTML preview). 4. Jalali dates only as text (`|fa`). 5. Viewer download links live in one server's

memory. 6. LibreOffice/Excel Online/Google Sheets were not tested (desktop Excel and IIS Express/.NET Framework 4.8 were). 7. xlsx import: charts/PivotTables/tables are not imported; shared formulas come in as current values. 8. The designer UI is optimised for desktop (not touch).

Not available (by design of the grid engine): *sub-reports as separate templates* - use nested bands plus several data sets (they cover master/detail/detail); free-positioned report objects outside the cell grid.

Also done: multi-level groups with SUBTOTAL, calculated columns, Jalali calendar functions, PDF password/permissions.

Next: OData/GraphQL helpers, chart import, template versioning, NuGet publishing, streaming for totals above the table, combo charts, EAN-13/Code 39/PDF417 barcodes.

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